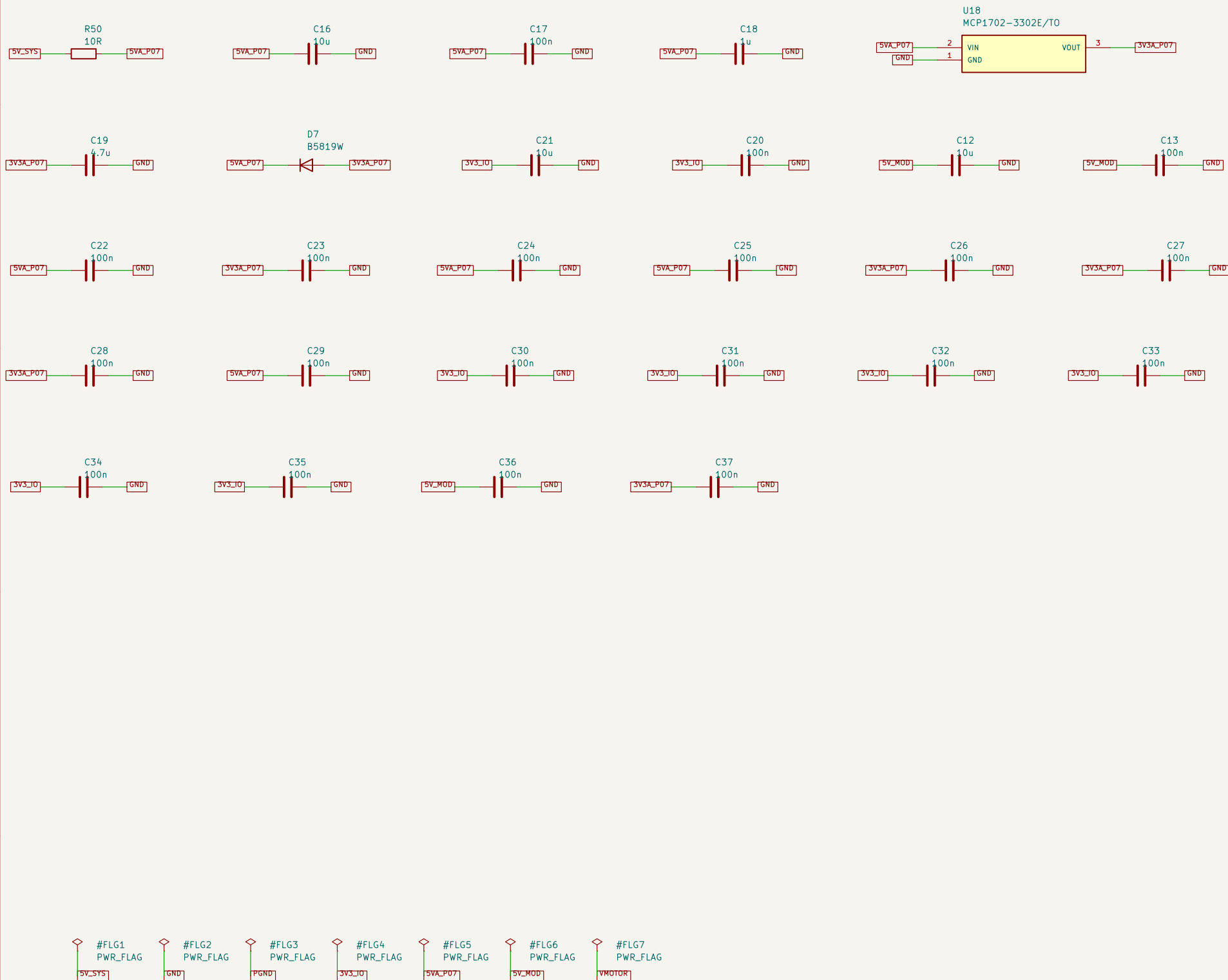


P07-S1 / 01 ZASILANIE LOKALNE I ARKUSZE

J_BP2: 5V_SYS (piny 10/12) i 3V3_I0 (pin 14) z P02 R4 przez P12. Logika (U10-U15) na 3V3_I0 – ta sama szyna co P04 R3. Analog na lokalnych 5VA_P07 (R50 10R) i 3V3A_P07 (U18).

5V_MOD = 5V_SYS za PTC F1: VCC modulu IBT-2 (74HC244) i bufora U16. GND (logika, analog) i PGND (prąd silnika) NIE są połączone na P07: wspólny punkt mas = P02 R4 J2 (VMOTOR).



MOC

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ANA

Plik: ANA.kicad_sch

DIG

Plik: DIG.kicad_sch

LOGIKA

Plik: LOGIKA.kicad_sch

MODUL

Plik: MODUL.kicad_sch

ZLACZA

Plik: ZLACZA.kicad_sch

SERWIS

Plik: SERWIS.kicad_sch

Tester drive carrier for an off-stack IBT-2 module; format S1 (2/3, level 5, S2-S3); schematic only.

Prototype / schematic + PCB review

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Title: EGRLab P07 DRIVE (IBT-2 / 2 x BTS7960B)

Size: A3	Date: 2026-10-05
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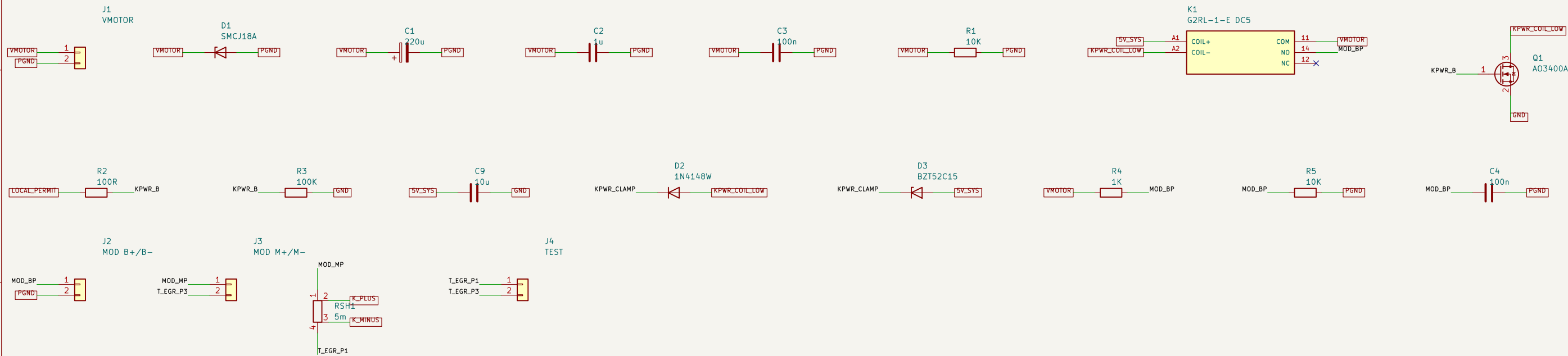
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P07-S1 / 02 VMOTOR, KPWR, MODUL B+/B-, BOCZNIK, PORT TEST

VMOTOR (P02 R4 J2, za F1 MINI 7,5 A) -> J1 -> KPWR K1 (NO) -> MOD_BP -> J2 -> modul B+. R4 1k lądzie 330 uF modulu przy otwartym KPWR; KPWR: cewka 5 V z 5V_SYS (decyzja 5.10). Q1 A03400A (zróżdło na GND) z LOCAL_PERMIT; clamp D2 + D3 15 V do 5V_SYS.

Modul M+ -> J3.1 -> RSH1 5 mOhm (Kelvin K_PLUS/K_MINUS) -> T_EGR_P1 -> J4.1 -> port TEST (P11). M- -> J3.2 = T_EGR_P3 -> J4.2. Prąd dodatni: kierunek A (RPWM, M+ wyżej).

PGND: powrót przez J1.2 do P02, B- modulu przez J2.2, TVS D1, C1-C4, R1/R5 (obwód cewki KPWR w domenie GND). Tor 10 A: pola ≥ 4 mm na obu warstwach (S1 3, jak P06 R2).

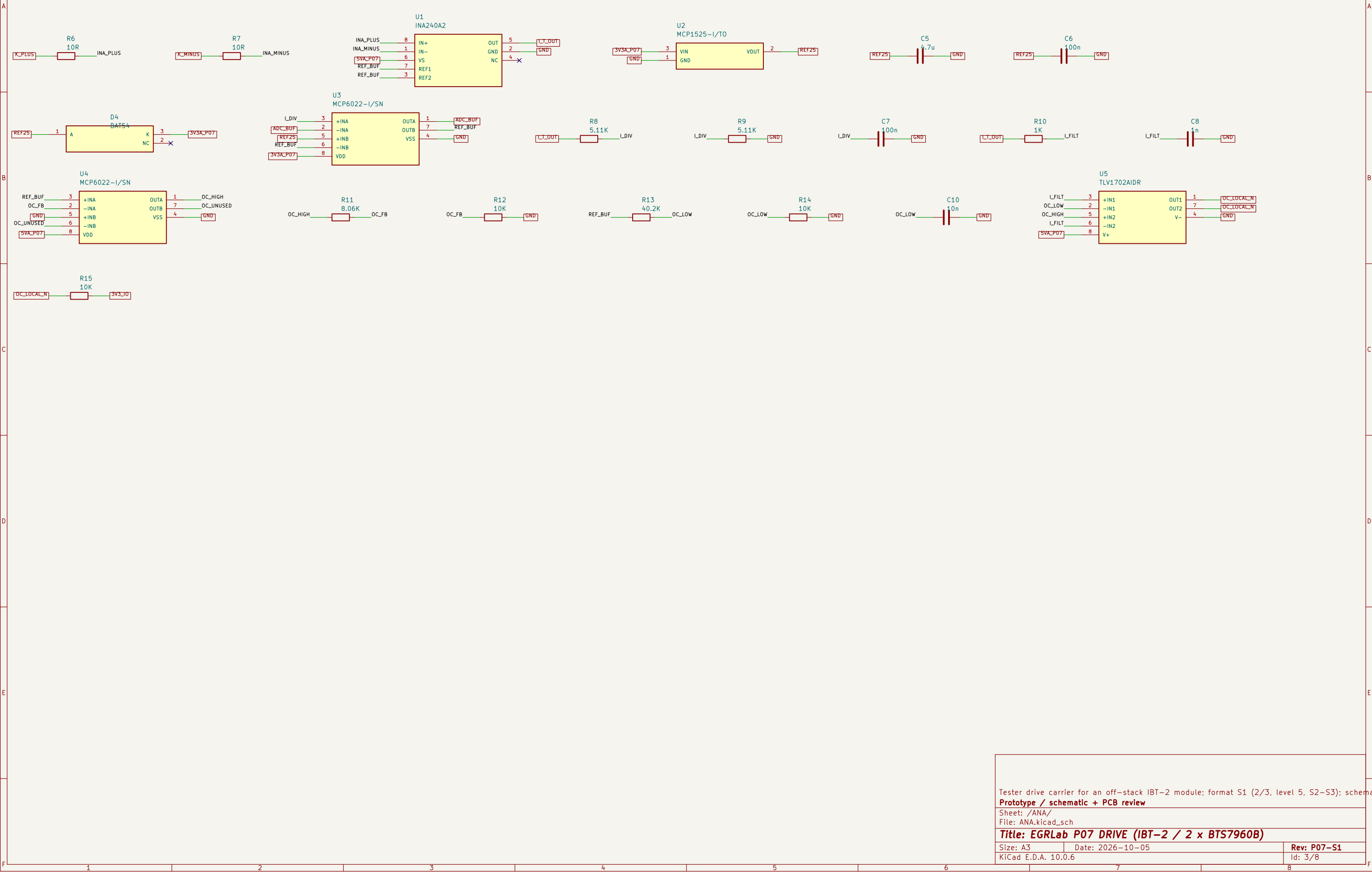


Tester drive carrier for an off-stack IBT-2 module; format S1 (2/3, level 5, S2-S3); schematic only. Prototype / schematic + PCB review			
Sheet: /MOC/ File: MOC.kicad_sch			
Title: EGRLab P07 DRIVE (IBT-2 / 2 x BTS7960B)			
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P07-S1 / 03 INA240, ODNIESIENIE, ITEST, OKNO OC

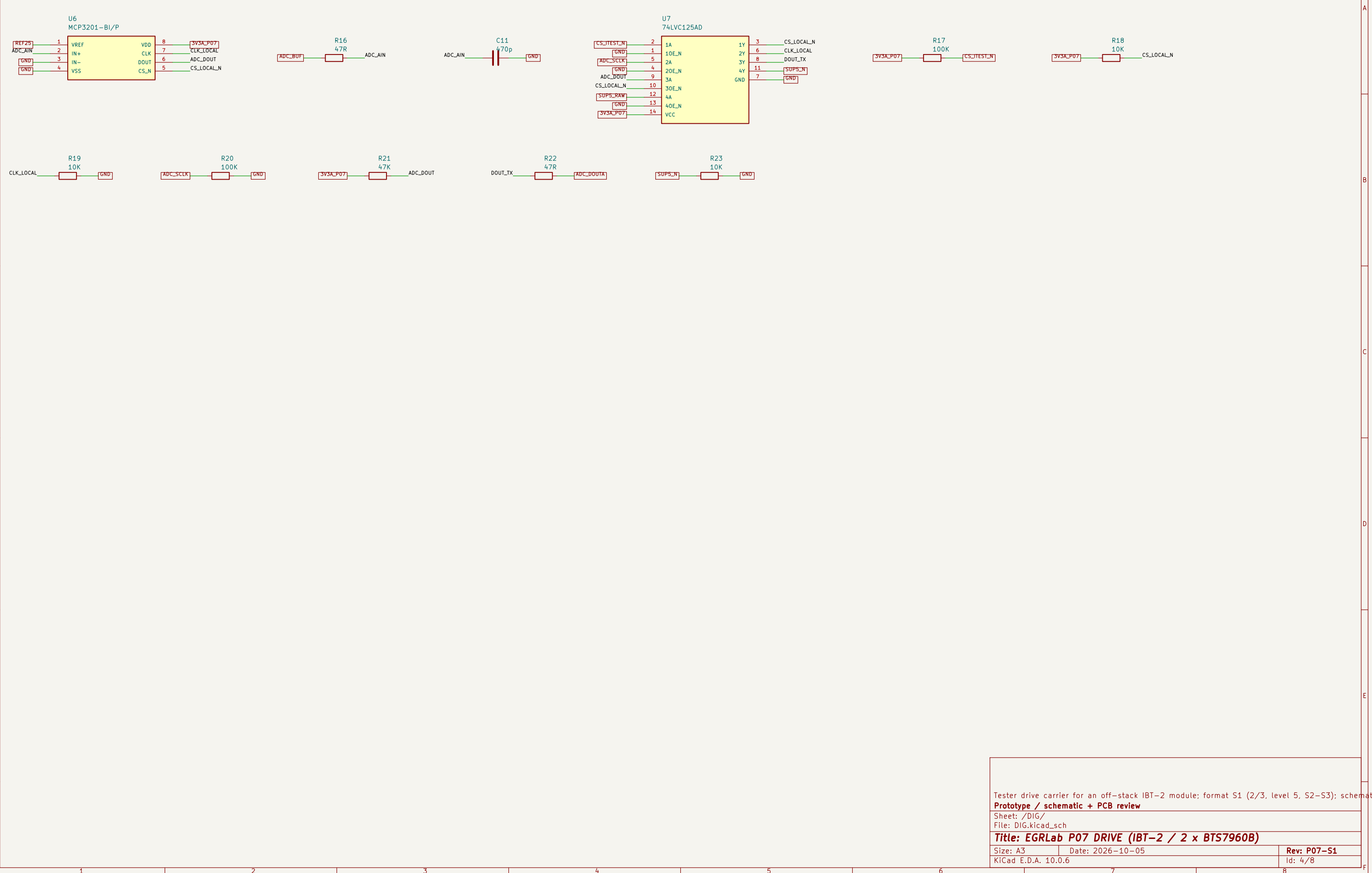
$L_T_{OUT} = REF_BUF + 50 \times 5 \text{ mOhm} \times I = 2,5 \text{ V} + 0,25 \text{ V/A}$. ITEST: R8/R9 1:2 + C7 100n (tau 0,26 ms) -> U3A -> MCP3201: 1,25 V + 0,125 V/A, +-10 A w 0..2,5 V.

Okno OC: L_FILT (R10 1k / C8 1n) vs OC_HIGH = REF_BUF x 1,806 (4,515 V, +8,06 A) i OC_LOW = REF_BUF x 0,1992 (0,498 V, -8,01 A), U5 open collector -> OC_LOCAL_N (R15 10k do 3V3_IO).



P07-S1 / 04 MCP3201 ITEST I BUFOR SPI

SPI mode 0, 16 zegarów, raw = (word >> 1) & 0x0FFF (jak P06 R2). DOUT_TX trojstanowy (OE = CS_LOCAL_N) + R22 47R na wspólnej ADC_DOUTA. U7 gate 4: SUP5_RAW (5 V) -> SUP5_N (3,3 V).

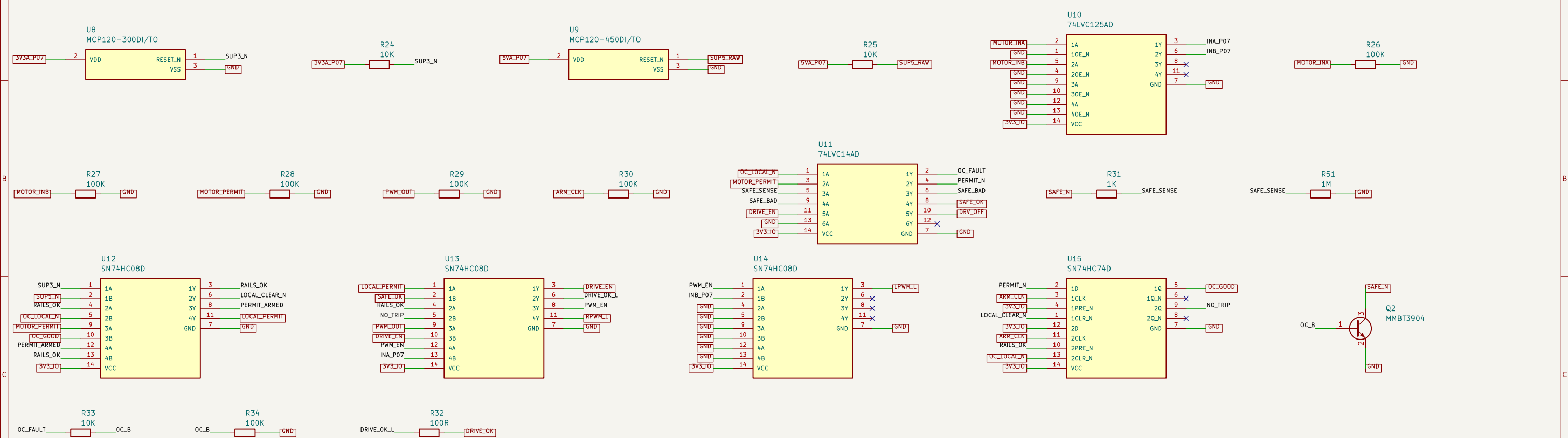


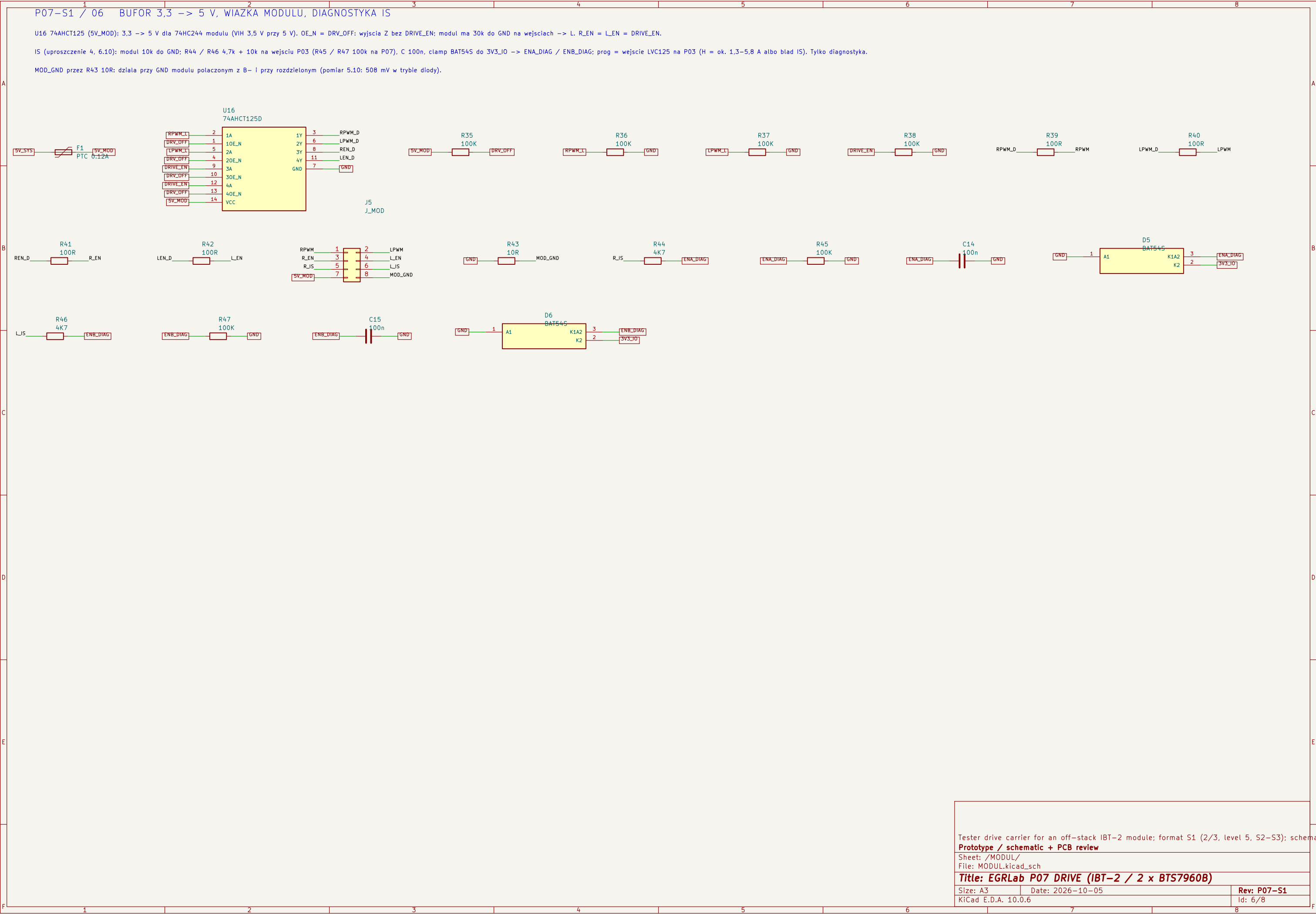
P07-S1 / 05 NADZORCY, ODBIORNIKI, BRAMKI, ZATRZASK OC, SAFE_N

RAILS_OK = SUP3_N & SUP5_N. LOCAL_CLEAR_N = RAILS_OK & OC_LOCAL_N. OC_GOOD: kasowany przez OC/zasilanie, ustawiany TYLKO zboczem ARM_CLK przy MOTOR_PERMIT = L (D = PERMIT_N).

LOCAL_PERMIT = MOTOR_PERMIT & OC_GOOD & RAILS_OK (-> KPWR). DRIVE_EN = LOCAL_PERMIT & SAFE_OK. RPWM = PWM_OUT & DRIVE_EN & MOTOR_INA; LPWM = PWM_OUT & DRIVE_EN & MOTOR_INB.

NO_TRIP (FF2): PRE = RAILS_OK (start = brak zadziałania), CLR = OC_LOCAL_N, D = 1 przy ARM_CLK. DRIVE_OK = RAILS_OK & NO_TRIP. Q2 sciąga SAFE_N podczas OC.

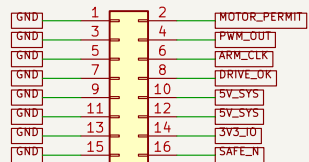
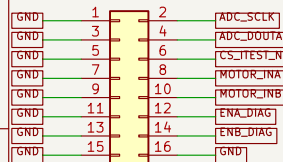




P07-S1 / 07 J_BP1 / J_BP2 DO P12 (KRAWEDZ A)

J_BP1 (S2, x płytki 26,5 / stosu 80,0): ADC_SCLK 2 / ADC_DOUTA 4 jak J_BP2 P03 R6, P05 R3 i J_BP P06 R2; CS_ITEST_N, MOTOR_INA/INB, ENA/ENB_DIAG do P03 R6.

J_BP2 (S3, 4, 5) 80,0 / stosu 133,5); MOTOR_PERMIT 2, PWM_OUT2 4, ARM_CLK 6, DRIVE_OK 8, SAFE_N 16 na tych samych pinach co J_BP2 P04 R3; 5V_SYS 10/12, 3V3_IO 14. Nieparzyste GND.



Tester drive carrier for an off-stack IBT-2 module; format S1 (2/3, level 5, S2-S3); schematic only. Prototype / schematic + PCB review	
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Title: EGRLab P07 DRIVE (IBT-2 / 2 x BTS7960B)

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